

Nikhilesh Vijay Chirutala

Full-Stack Software Engineer

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SUMMARY

Full-Stack Software Engineer with 2.8+ years of experience building and maintaining production-grade web applications across AI, CRM, trading, and cloud contact center platforms. Strong experience in Python, FastAPI, JavaScript, React.js, Next.js, REST APIs, MongoDB, AWS, and cloud-based application development. Experienced in designing backend services, developing responsive frontend interfaces, integrating third-party APIs, implementing authorization systems, automating production workflows, troubleshooting complex issues, and supporting applications throughout the software development lifecycle. Proven ability to work across backend, frontend, cloud, testing, deployment, and production support while delivering reliable and scalable software solutions.

TECHNICAL SKILLS

Languages: Python, JavaScript (ES6+), TypeScript, SQL, HTML5, CSS3

Frontend: React.js, Next.js, Angular, Tailwind CSS, Bootstrap, Jinja2

Backend: FastAPI, Node.js, Express.js, REST APIs, API Development, Backend Automation, Authentication, RBAC, ACL

AI & LLM: RAG, LLM Integration, Vector Search, Prompt Engineering

Databases: MongoDB, Neon DB, SQL

Cloud & DevOps: AWS, Amazon S3, Vercel, GitHub Actions, CI/CD, Linux

APIs & Integrations: Twilio, GitHub API, REST APIs, Postman

Testing & Tools: Postman, Jest, Unit Testing, Git, GitHub

WORK EXPERIENCE

Software Engineer • CA One Tech Cloud Inc.

Apr 2026 – Present | Bengaluru, India

- Developing backend services and automation for the Convoso cloud contact center platform, supporting large-scale call recording, cloud storage, metadata synchronization, and recovery workflows processing 10,000+ call recordings monthly.
- Designed and implemented a call-recording recovery and S3 storage management system from the ground up where no prior automation existed, reducing manual intervention and improving reliability of production recording workflows.
- Built backend automation to identify, recover, organize, and synchronize call recordings across storage systems while maintaining associated metadata and ensuring consistency between application records and cloud storage.
- Worked extensively with Python and AWS services to develop reliable cloud-based workflows for recording management, data processing, storage operations, and production support.
- Investigated and resolved production issues across the recording pipeline by tracing failures through backend services, storage workflows, metadata synchronization, and deployment environments.
- Performed deployment validation, functional testing, bug fixes, feature enhancements, and production troubleshooting to ensure backend changes were stable before and after releases.
- Collaborated with development and testing teams to understand production requirements, identify root causes, implement fixes, and continuously improve recording storage and recovery processes.

Key Projects:

CONVOSO — Cloud Contact Center Platform

Python, FastAPI, AWS, Amazon S3, REST APIs

- Developed and enhanced Python-based backend services for the Convoso cloud contact center platform, supporting lead routing, call recording management, data processing, cloud storage, and automated operational workflows.
- Designed recording-recovery workflows capable of processing 10,000+ recordings per month and restoring missing or inconsistent recording data.
- Implemented S3-based storage management workflows for handling recording files, validating storage state, and maintaining consistency between application metadata and cloud objects.
- Developed metadata synchronization processes to ensure recording information remained consistent across application systems and cloud storage.
- Automated repetitive recording-management and recovery tasks, reducing dependency on manual intervention and improving operational efficiency.
- Implemented backend validation and error-handling workflows to identify failed recording operations, handle unexpected storage states, and improve reliability of recovery processes.
- Worked with REST APIs and backend services to integrate recording workflows with existing contact center application components and data-processing pipelines.
- Investigated production issues by analyzing application behavior, backend workflows, storage states, and data inconsistencies to identify root causes and implement corrective fixes.
- Performed testing and deployment validation for backend changes, verifying recording workflows and storage operations across development and production environments.

- Worked as a full-stack software engineer across four production applications, contributing to architecture, feature development, backend services, frontend interfaces, testing, deployment, debugging, and production support.
- Designed and developed scalable backend modules using Python, FastAPI, REST APIs, and MongoDB while implementing application business logic and reusable service components.
- Developed secure REST APIs with authentication, role-based access control, and ACL-based authorization to enforce application-level permissions across users, teams, and resources.
- Built frontend functionality using React.js, Next.js, Angular, JavaScript, TypeScript, HTML, and CSS, integrating frontend components with backend APIs and application services.
- Integrated third-party services and APIs including Twilio and GitHub API to extend application functionality and automate application workflows.
- Managed application deployments using AWS and Vercel, maintained GitHub Actions CI/CD workflows, and supported Linux-based development and deployment environments.
- Performed API testing and functional validation using Postman, investigated application defects, implemented fixes, and supported releases across development and production environments.
- Worked in Agile development teams, participating in requirement analysis, implementation, testing, debugging, deployment, and continuous improvement throughout the software development lifecycle.

Key Projects:

AI Bull — AI-Powered Trading Platform *React.js, Next.js, FastAPI, Python, MongoDB, RAG, LLM*

- Developed an AI-powered trading and investment analysis platform providing tools for stock and mutual fund analysis, screening, and back-testing.
- Developed a Retrieval-Augmented Generation (RAG) based chatbot that retrieved relevant financial and application-specific information to provide contextual responses to user queries.
- Built backend APIs and business logic using FastAPI and Python to process financial data and support application workflows.
- Developed back-testing engines allowing users to evaluate investment strategies against historical market data.
- Implemented dynamic screening and filtering capabilities for stocks and mutual funds based on configurable financial parameters.
- Developed frontend functionality using React.js and Next.js, integrating interactive financial analysis features with backend APIs.
- Integrated the RAG chatbot with the application workflow to enable users to interact with relevant financial information through a conversational interface.

CRM.io — AI-Driven CRM System *FastAPI, Python, Jinja2, MongoDB, ACL, Twilio, REST APIs*

- Developed core CRM modules for managing Contacts, Tasks, communication workflows, and telephony operations.
- Built backend APIs and business logic using FastAPI and Python to support CRM workflows, user operations, and application data.
- Implemented role-based access control and ACL authorization to restrict CRM resources and operations based on user permissions.
- Integrated Twilio APIs to support telephony and communication functionality within the CRM platform.
- Developed server-rendered application interfaces using Jinja2 and integrated frontend workflows with backend services.
- Tested APIs and application workflows using Postman and contributed to debugging, bug fixing, and production support.

PERSONAL PROJECTS

JS Studio — Online JavaScript Playground *Next.js, GitHub API, Vercel*

- Developed an online JavaScript playground allowing users to write, execute, and experiment with JavaScript directly through a web interface.
- Integrated GitHub APIs to support project and repository-related workflows.
- Built and deployed the application using Next.js and Vercel.

AnyScrum — Project Management Platform *Next.js, Neon DB, Clerk, Vercel*

- Developed a project management platform for organizing projects, tasks, and team workflows.
- Implemented application authentication and user management using Clerk.
- Integrated Neon DB for persistent application data and deployed the platform using Vercel.

Guider — Interactive Learning Portal *Angular 19, TypeScript, SSR, Vercel*

- Built an interactive learning platform using Angular 19 and TypeScript with server-side rendering.
- Designed reusable frontend components and structured learning workflows for an improved user experience.
- Deployed the application on Vercel with a production-ready frontend architecture.

Contact Management App — Contacts Storage Application *React.js, JavaScript, Netlify*

- Developed a React-based contact management application for creating, viewing, updating, and managing contact information.
- Built reusable UI components and client-side application logic for managing contact workflows.
- Deployed the application on Netlify.

EDUCATION

B.Tech — Electrical & Electronics Engineering *2019 – 2023 | GPA: 7.73 / 10*

CERTIFICATIONS

- Agile Methodology Virtual Experience Program — Cognizant (Oct 2022)
- Python Programming Online Training — APSSDC (May–Jun 2021)